

# Nimble RX

Proposal | Nimble @ WashU Robotics, Fall 2026

## Problem

Prescription handling is largely manual: pharmacy staff count, label, and verify orders by hand, and patients/caregivers sort weekly medications into organizers, a repetitive, human error-prone task. Nimble RX explores a natural-language-initiated prescription handling pipeline built on Vision-Language-Action (VLA) models to fix the human risk of misplacement.

## Solution

### Pipeline

```
"Put the red prescription in Container A."  
-> Whisper STT -> schedule parser (local Qwen LLM)  
-> [(rx_i, container=i), ...]  
-> Kokoro TTS read-back confirmation  
-> for each (rx, container):  
    -> VLA Inf. #1: pick -> collision check: pass -> continue | fail -> withhold  
    -> VLA Inf. #2: place -> collision check: pass -> continue | fail -> withhold  
-> YOLO CV model: correctness check -> pass -> done | fail -> withhold
```

### Implementation

- **Data collection.** LeRobotDataset episodes (instruction, camera frame, 6-DOF grasp/place chunk) per item, initial set comprises 10 episodes/type, scaling toward  $\geq 50$ /type. Isaac Lab may accelerate collection.
- **Safety.** A YOLO CV model checks correct-container placement, serving as a post-placement evaluator and safety outcome predictor.
- **Training.** SFT on NORA 1.5 [1] via its official fine-tuning script, trained/evaluated per-item to keep the success signal dense. *Stretch:* DPO post-training, pairing successful vs. failed rollouts of the same instruction using CV verdicts as preference labels.

### Equipment

|                |                             |                  |                                            |
|----------------|-----------------------------|------------------|--------------------------------------------|
| <b>Arm</b>     | WidowX AI Base (6-DOF)      | <b>Organizer</b> | Containers for related prescription items  |
| <b>Gripper</b> | WidowX squeeze gripper      | <b>Items</b>     | 10 3D-printed prescription replicas        |
| <b>Cameras</b> | Intel RealSense D435i       | <b>Compute</b>   | RoboServer GPU inference                   |
| <b>Base</b>    | Linear actuator (workspace) | <b>Env.</b>      | Fixed, well-lit, curtained safety boundary |

### Key Deliverables

- **LeRobotDataset** of synchronized frames + user input + grasp/place action chunks, a **fine-tuned VLA** (NORA 1.5) for prescription pick-and-place, a **linear actuator** extending the arm's workspace, a **YOLO CV model** for safety and potentially a DPO reward, finally a **public release** of dataset, model, metrics, videos, and report on HuggingFace / GitHub.

### Timeline

|                  |                                                                                 |
|------------------|---------------------------------------------------------------------------------|
| <b>Wks 1–3</b>   | Equipment sourcing (3D printing, ordering), environment setup, RoboServer setup |
| <b>Wks 4–7</b>   | Per-item data collection, dataset build, preprocessing pipeline.                |
| <b>Wks 8–10</b>  | Fine-tuning, CV model, safety guardrails.                                       |
| <b>Wks 11–12</b> | Evals, documentation, public release.                                           |

### Success Metrics

- **Per-item success rate:**  $\geq 80\%$ , measured over 5 attempts per item type.
- **Safety effectiveness:**  $\geq 80\%$  of unsafe predicted actions correctly withheld.

### References

[1] Declare Lab. *NORA 1.5 Model Card*. Hugging Face, 2025.